

A Deep Learning Reconstruction Technique and Workflow to Enhance 3D X-ray Imaging Resolution and Speed for Electronics Package Failure Analysis

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Abstract— 3D X-ray imaging plays an important role in package level failure analysis. Like most other microscopies, X-ray microscopes (XRM) generally have field of view (FOV) limits for high-resolution imaging. As precise fault isolation becomes more challenging in large and complex IC packages, acquiring numerous high-resolution images to search for defects in a large FOV is required. For example, if a suspect fault region is 785 mm³ in a cylinder with 10mm in diameter and 10mm in height, more than 125 high-resolution scans with 6.28 mm³ each are required to cover the volume. This significantly diminishes XRM imaging throughput. In this report, we propose a new deep learning reconstruction method to address the issue of achieving high-resolution at large FOVs. This AI powered technique and workflow can be used to restore the resolution from a large FOV scan.

Keywords— X-ray, Deep-learning, Packages, Failure Analysis

I. INTRODUCTION

3D XRM imaging technique has been widely adopted to failure analysis and characterization of electronics systems and packages. With its non-destructive and high resolution imaging capability, deeply buried defects can be visualized prior to physical failure analysis (PFA). The X-ray microscopic technique, however, faces a challenge in imaging new IC packages where more densely packed interconnects are built in larger platforms. Because of the difficulty to pinpoint defect locations, it is often required to scan a large FOV with high resolution and speed, leading to the time-consuming and sometimes impractical X-ray imaging process.

In this paper, we introduce a novel deep learning based reconstruction method, capable of learning the point spread function (PSF) of low resolution data by training on a pair of high and low resolution data. By operationalizing the trained network model, low resolution data is transformed to high resolution data through an inference process. We will demonstrate on several IC packages that the proposed workflow can be utilized in fault isolation for a large area, in which standard XRM has not been practically possible due to the trade-off of image resolution and FOV, a fundamental limit for most microscopes.

II. METHOD

A conventional approach to improve image sharpness is to deconvolve the estimated PSF of the imaging system from

actual tomographic images, e.g. using Richardson-Lucy deconvolution [1-2]. However, the accurate derivation of PSF requires the analytical modelling of data acquisition geometry, which may not reflect actual measurement conditions of a given sample. Additionally, the deconvolution process is computationally expensive, often leading to high levels of noise and artefacts. In this paper, we propose a novel approach to recover image resolution where the deconvolution step is replaced with applying a convolutional neural network trained specifically on the spatially registered low-to-high resolution feature map (Fig. 1). The workflow starts with a spatial registration of multi-resolution data pair acquired for a sample. A network is then trained to remove the PSF effects at the low-resolution data. The recovery of image sharpness and removal of the uncorrelated image noise and artefacts are achieved because the network effectively learns the coupled effects of a finite PSF and partial volume artifacts.

In our first case study, a smartphone board with an A12 bionic System on Chip (SoC) package was commercially purchased (Fig. 2). The sample was tested under a JEDEC reliability test standard with 1,000 temperature cycles to create solder cracks. In the second case study, an A14 SoC package was acquired through a commercial channel. We focused the deep learning reconstruction study on the small redistribution layers and PCB metal traces. Training data pairs can be easily obtained with an XRM instrument because it is capable of acquiring multi-level resolution data.

III. RESULTS AND DISCUSSIONS

The low-resolution data at 10 $\mu\text{m}/\text{voxel}$ (Fig. 2b-c) and the high-resolution data at 2.1 $\mu\text{m}/\text{voxel}$ were acquired within the volume of the low-resolution data (Fig. 2b). First, these two datasets were spatially registered by using the reported methods [3-4] with 9-degree freedom to absorb small offsets resulting from imperfect machine alignments and possible sample drifts. Second, the resolution recovery network was trained using the modified image-to-image regression techniques [5-6] with loss functions, network structures and data augmentations tailored to the 3D tomographies. Finally, as a network model training is complete, the inference step was performed across the entire LFOV image or specific regions of interest, resulting in the recovered high-resolution image (Fig. 2d).

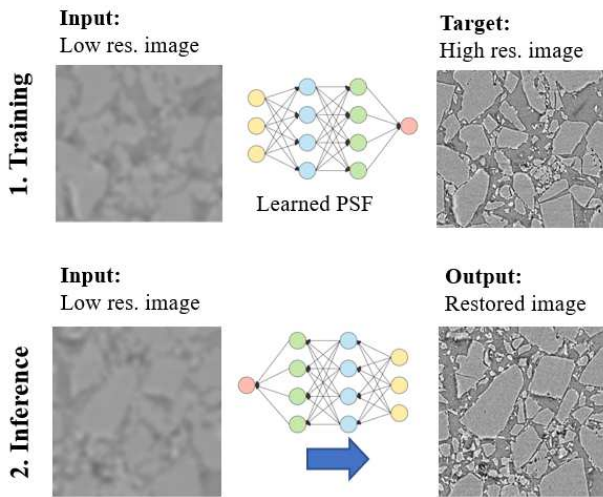


Fig. 1: the schematics of the deep learning reconstruction method to restore image resolution from the low-resolution image input.

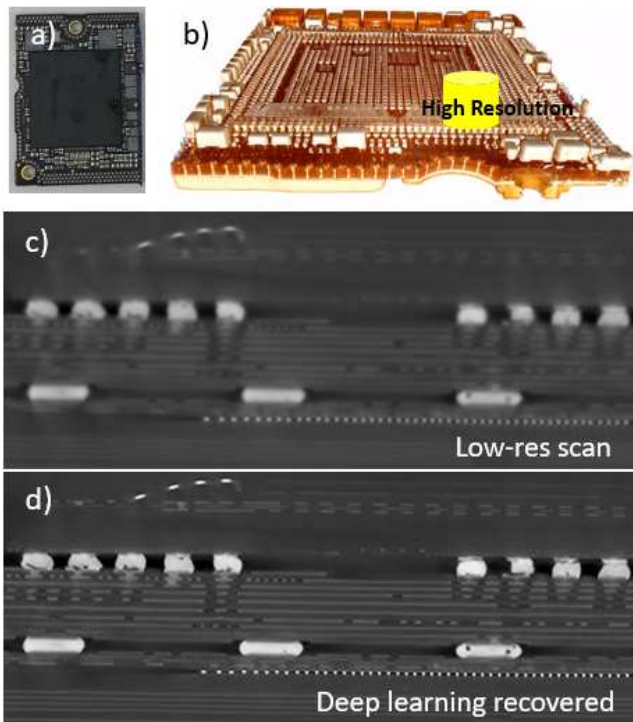


Fig. 2: a) A12 SoC package test vehicle; b) 3D color-rendered image of the low-resolution scan; c) a virtual cross-section of the acquired low-resolution image; d) the same virtual slice from the low-resolution scan but restored with the deep learning reconstruction method.

Fig. 2d shows the example slice of the restored image. In contrast to the original low-resolution image in Fig. 2c, the deep learning recovered image reveals more details of the bump cracks, while the same defect looked more ambiguous and less conclusive in Fig. 2c because of the resolution limit. Additionally, the beam hardening artifacts associated with the wire bonding are significantly reduced in the deep learning recovered image. The volume of the original low-resolution image is $\sim 14,130 \text{ mm}^3$ and the volume of the high-resolution image is $\sim 131 \text{ mm}^3$. The calculated volume gain is 108, the minimum number of scans without considering the overlapping regions would be required to achieve the volume and resolution of the deep learning recovered image.

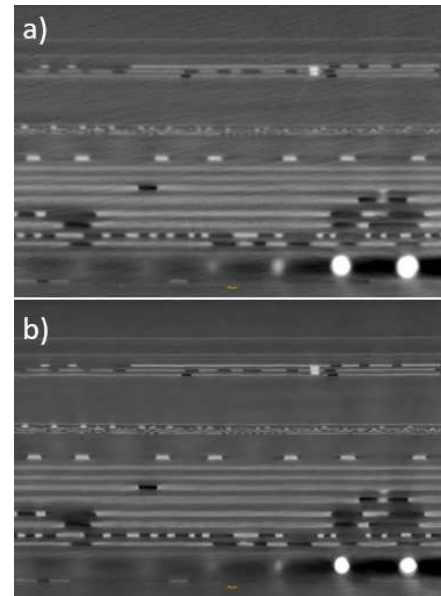


Fig. 3. a) the low-resolution image at $4.5 \mu\text{m/voxel}$; b) the image recovered by the deep learning reconstruction.

A pair of low-resolution and high-resolution data was acquired with a 4X objective lens at $4.5 \mu\text{m/voxel}$ and $1.5 \mu\text{m/voxel}$ respectively. The deep learning network was generated through training these two data. The reconstructed slice in Fig. 3b shows the image recovered by the deep learning reconstruction, revealing more microscopic details on the redistribution layers and PCB metal traces, compared with the original low-resolution scan in Fig. 3a. This example demonstrates that the deep learning reconstruction can work on the data with small structures.

IV. CONCLUSIONS

As new package architectures trend towards increased 3D packaging and heterogeneous integration, the traditional fault isolation techniques, such as acoustic microscopy, lock-in thermography, and time delay reflectometry, become less effective to localize the true fault regions. The deep learning powered X-ray imaging technique proposed here can extend to fault isolation applications, which are traditionally impractical, because it can be used to recover the resolution of a large volume.

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